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Probabilistic forecasting of renewable curtailment under regime change: an open dataset and a conformal method with distribution-shift-aware coverage control for Chile's National Electricity System

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Probabilistic forecasting of renewable curtailment under regime change: an open dataset and a conformal method with distribution-shift-aware coverage control for Chile's National Electricity System

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Abstract

The curtailment of variable renewable generation has become a first-order operational and economic problem in power systems with rapid solar and wind expansion. Chile's National Electricity System (SEN) curtailed roughly 19,000 GWh of renewable energy between 2022 and mid-2026, and the large-scale deployment of battery energy storage during 2025 altered the statistical behaviour of curtailment in a way that breaks the assumptions of standard forecasting practice. This paper makes two contributions. First, we release an open, plant-level, errata-corrected dataset of daily and hourly curtailment for the entire SEN, covering 53 continuous months and 300 plants across four technologies, whose annual totals reconcile exactly with the official system operator figures. Second, using this dataset, we develop a probabilistic seven-day-ahead curtailment forecaster and pair it with a conformal calibration layer designed to remain informative under distribution shift. We show that point forecasting offers little marginal value for this problem, since a seasonal-naïve baseline is difficult to beat in absolute error, and that the operationally relevant challenge is therefore reliable uncertainty quantification rather than point accuracy. We further show that the 2025 storage-driven regime change degrades naïve conformal calibration not by breaking coverage but by inflating it: the static split remains over-covered through the transition and its intervals widen substantially. On a synthetic panel that isolates covariate shift, the textbook weighted-conformal remedy fails by returning uninformative infinite intervals; on the real data, an adaptive scheme that combines a probability-integral-transform score with online recalibration and a transport of the calibration scores restores sharpness while holding coverage near the nominal level through the ramp, narrowing the transition intervals by about a third. The static split provides finite-sample marginal coverage under exchangeability, and the adaptive layer provides long-run coverage control when exchangeability fails. The dataset and code are publicly available.

Keywords: renewable curtailment, probabilistic forecasting, conformal prediction, distribution shift, battery energy storage, power system operations

1. Introduction

The share of variable renewable energy in electricity systems has grown faster than the transmission and demand able to absorb it, and curtailment, the deliberate reduction of available solar, wind or hydropower output, has become a routine consequence of that imbalance rather than an exceptional event, documented across systems worldwide [1]. In Chile's National Electricity System (Sistema Eléctrico Nacional, SEN), recorded curtailment rose from about 1.5 TWh in 2022 to more than 6 TWh per year by 2024, and the cumulative figure over the 2022 to mid-2026 period reaches roughly 19 TWh. For the owners of affected plants this is generated energy that is never remunerated, and for the system it is a growing signal of the mismatch between where and when renewable energy is produced and where and when it can be used.

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Anticipating curtailment is therefore of direct operational and commercial value. An asset owner who knows, days in advance, how much a given plant is likely to curtail can adjust maintenance, commercial and dispatch decisions accordingly. Yet curtailment is considerably harder to forecast than generation. Generation depends primarily on weather and installed capacity, which are well understood and increasingly predictable, whereas curtailment depends on the state of the whole system: net load, transmission congestion, the marginal cost of energy, and the behaviour of other assets. It is a downstream, system-level quantity, and this makes it both more difficult and less studied than generation forecasting.

Two features of the Chilean case make it a particularly informative setting. First, curtailment in the SEN is highly concentrated: a small number of regions in the north, where solar capacity has outpaced the north-to-south transmission corridor, account for the large majority of national curtailment. Second, and central to this paper, the system underwent a sharp structural change during 2025 with the large-scale entry of battery energy storage systems (BESS). Storage absorbs surplus energy that would otherwise be curtailed, and its deployment altered the statistical distribution of curtailment over a period of months. A forecasting model calibrated on the pre-storage regime is therefore miscalibrated on the post-storage regime, not because the model is wrong but because the system it describes has changed. This is a concrete, well-documented instance of the distribution-shift problem that pervades data-driven forecasting, and it is the methodological core of this work.

A further difficulty, often underappreciated in applied forecasting, is that a point forecast of curtailment is of limited use for decision-making. Curtailment is a zero-inflated, heavy-tailed quantity: on a large fraction of days a given plant curtails nothing, and on the days it does the amounts are highly variable. A single predicted number conveys neither the probability that curtailment will occur nor the range it might take. What a decision-maker needs is a calibrated statement of uncertainty, an interval that contains the realised value with a stated and reliable probability. Providing such intervals, and keeping them informative when the system changes regime, is the problem we address.

This paper makes two contributions. The first is a data contribution. We construct and release an open, plant-level dataset of curtailment in the SEN at daily and hourly resolution, spanning 53 continuous months and 300 plants across solar, wind and two hydropower technologies, built from the public reports of the system operator and reconciled against the official aggregate figures. In the course of building it we identified and corrected eight classes of errata in the original public files, using deterministic and reproducible rules, and recovered two calendar days that were missing from the operator’s own daily breakdown. To our knowledge no open curtailment dataset at this granularity exists for a Latin American power system. The second is a methodological contribution. Using this dataset, we build a two-part probabilistic forecaster for plant-level curtailment and wrap it in a conformal calibration layer scored by the probability integral transform. We characterise how the 2025 storage-driven regime change inflates standard calibration into over-covered, unnecessarily wide intervals, we show that the textbook remedy for covariate shift fails on a controlled synthetic panel by returning uninformative intervals, and we propose an adaptive scheme that restores interval sharpness through the transition while keeping coverage near nominal, distinguishing throughout between the finite-sample guarantee of the static split and the long-run guarantee of the adaptive layer.

The remainder of the paper is organised as follows. Section 2 reviews related work on curtailment forecasting and on conformal prediction under distribution shift. Section 3 describes the dataset and its construction. Section 4 presents the base forecaster and the conformal calibration layer. Section 5 reports the empirical results on real data. Section 6 discusses operational implications and threats to validity, and Section 7 concludes.

2. Related work

Curtailment forecasting has received far less attention than renewable generation forecasting, and existing studies have largely relied on point-forecast machine-learning models for individual grids, without addressing either distribution-free uncertainty quantification or structural regime change. The present work differs on both counts, and connects the applied problem to the literature on conformal prediction under distribution shift.

Conformal prediction constructs prediction sets with finite-sample, distribution-free coverage under exchangeability [2], the very assumption a regime change violates. Two lines of work relax it. Weighted conformal prediction extends the guarantee to covariate shift by reweighting calibration points according to a density ratio between the calibration and test distributions [3]; this presupposes that the conditional law $\mathbb{P}(Y | X)$ is invariant, which fails when the change is in the conditional itself. Conformal validity beyond exchangeability, with fixed non-uniform weights

and an explicit coverage-gap bound in total variation, covers the regime-change case directly [4], and online adaptive conformal inference attains long-run coverage under arbitrary shifts by updating the miscoverage level from observed errors [5]. Our adaptive scheme builds on these two, scores observations by their probability integral transform following the distributional-conformal construction of Chernozhukov et al. [8], and uses monotone one-dimensional optimal transport [7] to describe and correct the displacement of the score distribution across regimes. Proper scoring rules, in particular the continuous ranked probability score [6], provide the evaluation basis for the probabilistic forecasts.

3. Data

3.1. Source and scope

The dataset consolidates the curtailment records published monthly by Chile’s National Electricity Coordinator (Coordinador Eléctrico Nacional, CEN) into two validated relational tables. The daily table contains 293,678 records and the hourly table 6,906,761 records, both covering the continuous period from 1 January 2022 to 31 May 2026, that is 1,612 days built from all 53 monthly reports, with no missing days. Records span 300 generation plants across four technologies: solar photovoltaic (75 plants), wind (56), run-of-river hydropower (140) and reservoir hydropower (29). Twelve hydropower plants are labelled inconsistently across the operator’s monthly reports, alternating between the run-of-river and reservoir categories; the plant registry resolves each of them to a single canonical technology, so that a count of distinct plant-technology pairs in the raw daily table returns 312 rather than 300. Hydropower curtailment records are present from June 2024 onward, with hourly hydropower detail from July 2024, following the coverage of the underlying public reports.

Each plant is linked to a metadata registry matched to the CEN installation catalogue, with 18 fields including the official identifier, installed capacity, owner, region, province, municipality, geographic coordinates, commissioning date, connection point and regulatory classification; the matching method is recorded for every plant. Days and hours without curtailment are stored explicitly as zero-valued records, which represent 49.8% of daily and 83.6% of hourly rows. This explicit encoding of zeros is deliberate: it preserves the zero-inflated structure of the quantity, which is essential for the modelling in Section 4. Figure 1 shows the geographic distribution of the plants, with the concentration of solar capacity in the northern regions and of hydropower in the south.

3.2. Construction and validation

The 53 monthly reports were retrieved from the CEN public portal with a versioned download script, parsed and loaded into a relational database, and the original files were preserved unmodified with SHA-256 checksums. During construction we identified eight classes of errata in the original publications. Five were corrected or recovered deterministically in the curated tables and three are limitations of the source that we document without correction. Among the corrections, two omitted calendar days were recovered from the corresponding monthly hourly sheets and cross-checked to the millesimal against the source files’ own total columns; one of these was detectable only by a structural continuity check, since the omitted day carried zero energy and was therefore invisible to any energy-based validation.

Two independent checks establish the fidelity of the result. Annual totals from the daily table match the CEN official summary exactly, to a difference of 0.000 GWh, in every year from 2022 to 2026, and aggregate 2022 wind and solar curtailment reproduces the 1.47 TWh independently reported by Fraunhofer Chile for that year. At monthly resolution the agreement is close but not exact: in a census month-by-month reconciliation of hourly sums against daily totals over all 53 months, 50 months agree within 1% and 35 agree exactly, while the three remaining months correspond to documented inconsistencies within the operator’s own publications. The residual sub-percent differences arise from the operator reassigning the same monthly energy differently across sister run-of-river units between its monthly hourly sheets and its annual closing, with a net effect of zero at system level, which is the same mechanism as errata classes six and seven. The dataset totals 19,030 GWh of recorded curtailment over the period.

The dataset is publicly available on Zenodo under a CC BY 4.0 licence [10], together with the data dictionary, the errata log and the integrity checksums. The processing pipeline that regenerates every curated table is available in the companion code repository.

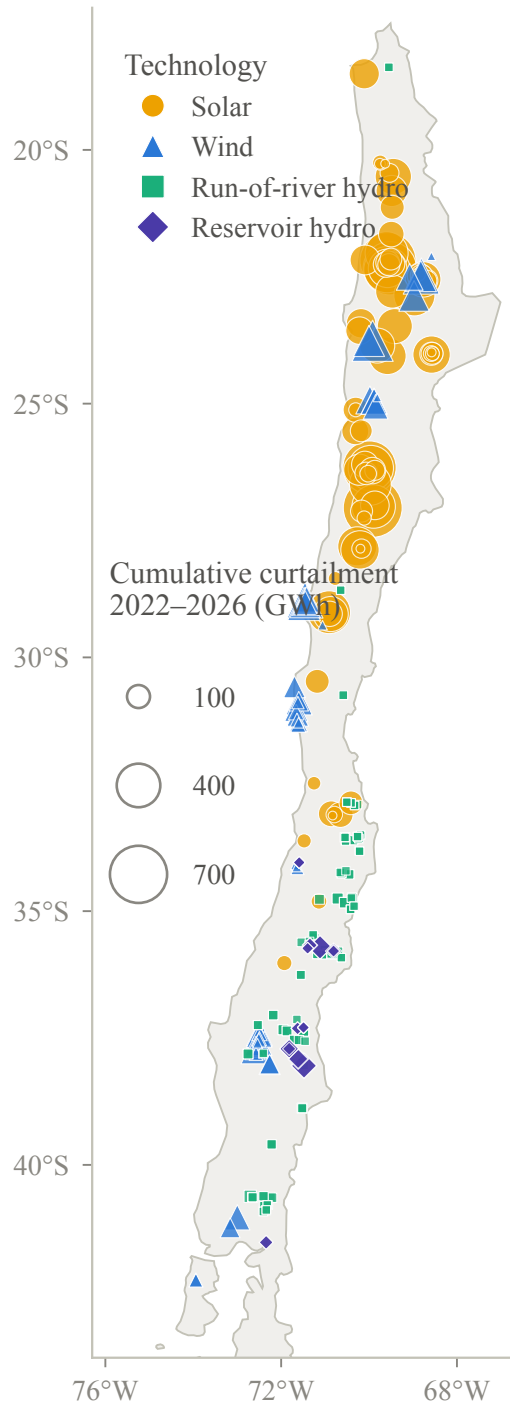


Figure 1: Geographic distribution of the 300 power plants in the dataset, coloured and shaped by technology, with marker area proportional to each plant's cumulative curtailed energy over January 2022 to May 2026.

3.3. Exploratory characterisation

Three features of the data govern the modelling choices that follow. First, curtailment is strongly zero-inflated and the occurrence of curtailment is persistent in time, with a daily lag-one autocorrelation of roughly 0.85 at system level and a marked weekly cycle. Second, the positive magnitudes are heavy-tailed and approximately lognormal, with a ratio between the 99th and 50th percentiles of the order of twenty-five across technologies, and of about fifteen for solar alone. Third, the 2025 storage deployment shifted the distribution of hourly curtailment magnitudes substantially while leaving the intraday shape and the occurrence probability almost unchanged, so that the regime change is felt in scale rather than in timing. Figure 2 illustrates the first two of these features. These three facts, zero inflation, heavy tails and a scale-type regime shift, motivate respectively the two-part model, the choice of evaluation metric and the adaptive calibration described next.

4. Methods

4.1. Problem setup and base forecaster

For each plant and each origin day t we predict curtailment at horizon $t + 7$. Because roughly half of plant-days carry no curtailment and the positive amounts are heavy-tailed, a single regression is ill-suited and we adopt a two-part (hurdle) model. Let x denote the feature vector available at the origin, built only from information observed up to t , comprising lagged curtailment, moving averages, a lagged occurrence indicator, calendar terms and static plant attributes. The first part is a classifier estimating the occurrence probability $p(x) = \mathbb{P}(Y > 0 \mid x)$. The second part models the positive magnitude as lognormal, $\log Y \mid Y > 0, x \sim \mathcal{N}(\mu(x), \sigma^2)$. The predictive law is the mixture

$$Y \mid x \sim (1 - p(x))\delta_0 + p(x)\text{LogNormal}(\mu(x), \sigma^2), \quad (1)$$

where δ_0 is a point mass at zero. Writing Φ for the standard normal CDF, the mixture CDF is $F_x(y) = 0$ for $y < 0$, has an atom $F_x(0) = 1 - p(x)$ at the origin (so that $F_x(0^-) = 0$), and for $y > 0$ equals

$$F_x(y) = (1 - p(x)) + p(x)\Phi\left(\frac{\log y - \mu(x)}{\sigma}\right).$$

Both parts are fitted with gradient boosting on data up to the end of 2023. The dispersion σ is estimated out of fold to avoid the downward bias of the in-sample residual variance of a boosted model; on the real data $\hat{\sigma} = 1.70$, which we return to in Section 5 because it sets the sharpness floor of the whole pipeline.

4.2. Conformal calibration with a PIT score

A point forecast, or even the raw mixture (1), does not by itself guarantee that a stated interval contains the realised value with the nominal probability. We obtain that guarantee by conformal calibration. Rather than measuring non-conformity in energy units, which are not comparable across plants of very different size, we score each observation by its probability integral transform (PIT) under the predictive law,

$$s = F_x(y), \quad (2)$$

following the distributional-conformal construction of Chernozhukov et al. [8]. Because F_x has an atom at zero, the score is randomised there in the standard way for mixed distributions [9]: for $y = 0$ we set $s = F_x(0^-) + U(F_x(0) - F_x(0^-)) = (1 - p(x))U$ with $U \sim \text{Uniform}(0, 1)$, while for $y > 0$ no randomisation is needed. Under the true predictive law s is then $\text{Uniform}(0, 1)$, and it lives on a common $[0, 1]$ scale across plants of any size, which is why it handles the atom and the plant heterogeneity cleanly.

This score has a further property that is decisive under regime change. For a positive observation the score is a monotone transform of the standardised log-magnitude $z = (\log y - \mu(x))/\sigma$, namely $s = (1 - p) + p\Phi(z)$. A change of scale of the magnitude distribution by a factor c is a location shift $\log c$ of $\log Y$, hence a translation of z by $\delta = \log c/\sigma$. It is therefore a *translation on the latent Gaussian (probit) scale*, and only a monotone displacement of the law of s on $[0, 1]$; it is not a translation of s itself. This low-dimensional structure, established under the empirically supported assumption that the storage-driven shift rescales the positive magnitudes while leaving the occurrence probability and

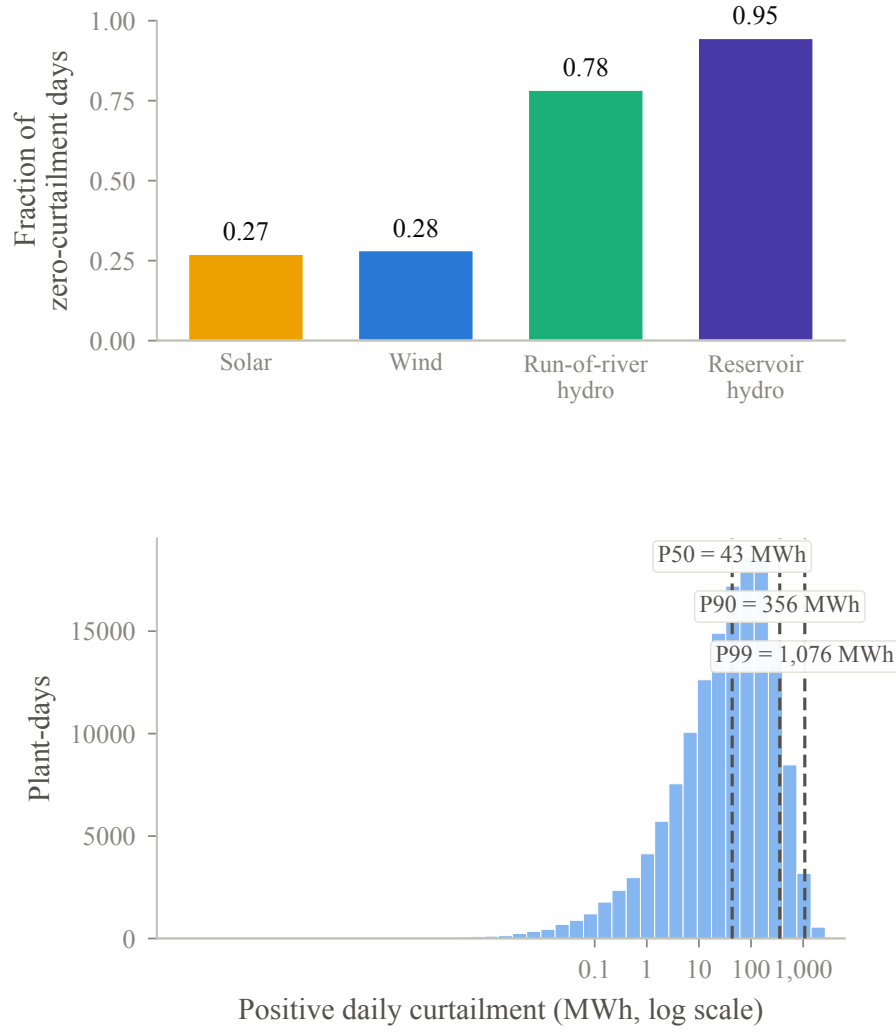


Figure 2: Distributional structure of plant-level curtailment. Top: share of zero-valued plant-days by technology. Bottom: distribution of positive magnitudes on a logarithmic scale, with the 50th, 90th and 99th percentiles marked. The combination of a large point mass at zero and a heavy right tail motivates the two-part model of Section 4.

the intraday timing essentially unchanged (Section 3), is what lets the displacement be estimated from a short recent window while the shape of the tail is inherited from the large calibration set.

Given a calibration set of size n scored by (2), the conformal quantile is the order statistic $\hat{q} = s_{(k)}$ with $k = \lceil (n+1)(1-\alpha) \rceil$, which yields the finite-sample marginal guarantee $\mathbb{P}(Y \leq U(x)) \geq 1-\alpha$ under exchangeability; when $k > n$ the quantile is $+\infty$ and the interval is the whole line. The upper prediction limit $U(x) = F_x^{-1}(\hat{q})$ is recovered by inverting the mixture: it is zero when $\hat{q} \leq 1-p(x)$ (the quantile falls in the atom), it is $+\infty$ when $\hat{q} \geq 1$, and otherwise

$$U(x) = \exp\left(\mu(x) + \sigma \Phi^{-1}\left(\frac{\hat{q} - (1-p(x))}{p(x)}\right)\right). \quad (3)$$

The two boundary cases are the two ends of the clipped argument $(\hat{q} - (1-p))/p \in [0, 1]$ in (3). A short proof of the inversion is given in Appendix A, and it was additionally checked numerically against oracle coverage.

4.3. Coverage under regime change

Exchangeability fails when the system changes regime, and the storage-driven shift of 2025 is such a failure. It fails, moreover, already within each regime: the daily systemic state is shared by all plants, so the calibration scores arrive in strongly correlated daily blocks, and the effective sample size for the fluctuation of coverage is of the order of the number of days, not of rows. For this reason every coverage figure we report carries a standard error clustered by date. We consider four responses to the shift.

Weighted conformal. Reweighting each calibration point by an estimate of the density ratio between calibration and test distributions is the standard remedy for *covariate* shift [3]. It is the wrong tool here: the storage ramp changes the conditional law $\mathbb{P}(Y | x)$, not the marginal of x , so no reweighting of old scores can reflect the new conditional, and the density-ratio estimate degenerates because features that are monotone in time separate calibration from test almost perfectly. On the synthetic panel, where covariate shift can be isolated, the per-point implementation of [3] returns infinite intervals for a large fraction of test points (Section 5); we report this as a negative result that motivates the alternatives, and we take the fixed-recency-weight scheme of Barber et al. [4], which bounds the coverage gap in total variation without a classifier, as the correct recency benchmark.

Sliding-window recalibration. We discard old calibration data and recalibrate on a trailing 60-day window refreshed every 7 days. Because the base model forecasts at a 7-day horizon, the label of an origin is realised only 7 days later; to avoid using information unavailable at the forecast origin, every online calibration window ends 7 days before its target (a horizon embargo). The static split does not use online calibration and is therefore unaffected by the embargo, which serves as an implementation control.

Adaptive conformal inference (ACI). We update the effective miscoverage level online [5], $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$ with $\text{err}_t = \mathbb{1}\{Y_t > U_t\}$, where the error feedback is delayed by the same 7-day embargo. This yields long-run average coverage without exchangeability, at the cost of the finite-sample guarantee of the static split.

Transport of scores. Motivated by the monotone-displacement structure above, we transport the calibration scores from the old regime onto the new one before conformalising. Let $\hat{F}_{\text{old}}$ and $\hat{F}_{\text{new}}$ be the empirical CDFs of the calibration scores and of the scores in the recent window; the monotone one-dimensional optimal-transport map is the increasing rearrangement $T = \hat{F}_{\text{new}}^{-1} \circ \hat{F}_{\text{old}}$, estimated nonparametrically by interpolation between empirical quantiles. In the upper tail, where the short window is noisy, the map is shrunk towards the identity in proportion to a bootstrap estimate of its variance, so that a noisy new quantile is replaced by the stable old one rather than extrapolated. Because the transport map is data-dependent it breaks the exchangeability of the transported scores with the test score; we therefore restore validity by wrapping the transported scores in the online adaptive scheme, so that coverage is controlled by the ACI update and the transport contributes only sharpness. An idealised validity argument, exact under an oracle map and controlled through the total-variation framework of [4] under an estimated map, is given in Appendix B.

4.4. Evaluation

We evaluate probabilistic forecasts by empirical coverage against the nominal level, by mean interval width over finite intervals (sharpness), by the fraction of infinite intervals, and by the continuous ranked probability score (CRPS), which is a property of the base predictive rather than of the calibration layer and is therefore reported once per period. Because all plants share the state of the system on a given day, the standard errors of every coverage figure are clustered by date; these are of the order of one percentage point, so that differences between methods smaller than a few points

Table 1: Mean absolute error (MWh) of the base forecasters on the out-of-sample period. Lower is better.

Model	2024	2025	2026	Total
Seasonal-naive (weekly)	84.3	86.3	84.2	85.1
Quantile GBM (median)	88.5	89.9	84.3	88.3
Persistence	88.4	96.3	90.7	92.2
Point XGBoost	92.7	94.7	89.7	93.0
Hurdle (mixture median)	104.4	104.8	101.6	104.1

are within noise. The calibration set is January to August 2024; the first test month, September 2024, is a stable pre-ramp regime, the storage ramp spans October to December 2024, and the remaining periods are taken by semester. The base forecaster is benchmarked against persistence, a seasonal-naive predictor, a point gradient-boosting model and a quantile gradient-boosting model.

5. Results

5.1. Point forecasting offers little marginal value

Table 1 reports the mean absolute error of the base models on the out-of-sample period. The seasonal-naive predictor, which simply averages the same weekday over recent weeks, is the most accurate of all, and is not beaten by the gradient-boosting models; the hurdle mixture median, which is optimised for the full predictive law rather than for absolute error, is the least accurate on this metric. This ordering is a direct consequence of the strong weekly persistence documented in Section 3: when a quantity is that persistent, sophisticated point models add little over a well-chosen calendar rule. The operational conclusion is that value does not lie in improving the predicted number, but in quantifying its uncertainty reliably, which is the object of the conformal layer.

5.2. Calibration under the regime change

On real data the storage shift does not break coverage; it inflates it. Under a static conformal split, empirical coverage stays between 89% and 95% in every period (Table 2), so the severe over-coverage seen on synthetic data is absent, because the base forecaster, through its lagged and moving-average features, absorbs much of the shift into its own predictive. The cost of the shift is therefore paid in width rather than in coverage, and it is concentrated in the ramp: the static split reaches 95.3% coverage with a mean width of 1203 MWh during October to December 2024. Figure 3 traces this behaviour over time: the static split drifts above the nominal level as the ramp advances, while the adaptive schemes track it more closely.

The adaptive schemes pay off precisely there. Transport of the calibration scores wrapped in online adaptation ($\gamma = 0.05$) brings the ramp intervals down to 806 MWh at 90.9% coverage, a reduction of about a third at a coverage within one point of nominal, and the sliding window gives 834 MWh at 92.0%. Outside the ramp the trade-off is ambiguous: with the static split already near nominal, the adaptive methods mostly exchange coverage for width and can under-cover, as in 2025-S1, and their gains are small and within the date-clustered standard error. Figure 4 makes this trade-off explicit across methods and periods. Two honest limitations frame the result. First, the base dispersion is large on real data ($\hat{\sigma} = 1.70$), so a substantial part of every width originates in the base model rather than in the calibration layer, which calibrates that dispersion rather than removing it; the sharpness floor is a property of the hurdle, not of the conformal step. Second, the weighted-conformal failure that motivates the adaptive schemes is established on the synthetic panel, where covariate shift can be isolated: the per-point implementation returns infinite intervals for the majority of test points during the ramp. It is a negative result about the tool, not about the data, and it is reported as such.

6. Discussion

The results carry a clear operational message. For plant-level curtailment at a seven-day horizon, effort is better spent on reliable uncertainty quantification than on chasing point accuracy, which a calendar baseline already delivers.

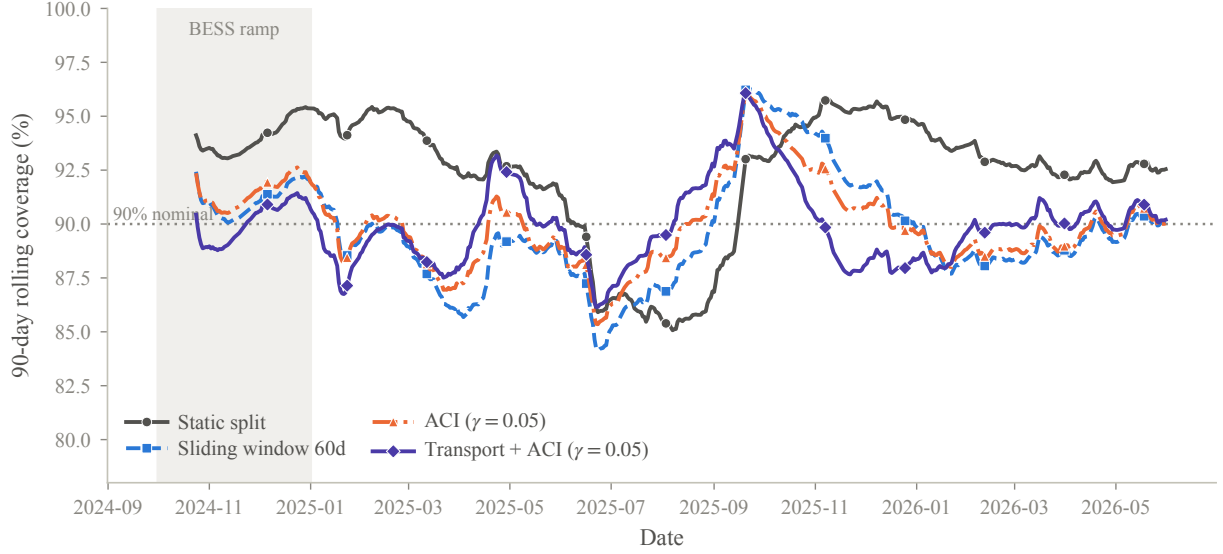


Figure 3: Rolling empirical coverage (90-day window) of the conformal methods on real data, computed with the 7-day horizon embargo. The dashed line marks the 90% nominal level and the shaded band marks the storage ramp of October to December 2024. The static split drifts into over-coverage as the ramp advances, whereas the adaptive schemes remain close to nominal.

Table 2: Coverage and mean interval width of the conformal methods by period, on real data, with the 7-day horizon embargo. Each cell shows coverage in percent (clustered standard error by date in parentheses) over mean interval width in MWh. Nominal coverage is 90%. The base-model CRPS is a property of the predictive, not of the calibration layer, and is reported once per period. Superscripts give the fraction of infinite intervals where nonzero.

Method	test_pre (Sep 24, stable)	test_ramp (Oct–Dec 24, ramp)	2025-S1	2025-S2	2026-S1
Static split (PIT)	93.5 (1.03) / 799	95.3 (0.45) / 1203	89.4 (1.04) / 455	94.0 (0.45) / 750	92.5 (0.58) / 535
Sliding window 60d	93.0 (1.04) / 742	92.0 (0.73) / 834	85.4 (1.16) / 323	92.9 (0.51) / 650	89.5 (0.76) / 419
ACI ($\gamma=0.02$)	93.0 (1.06) / 759	93.5 (0.55) / 915	85.7 (1.16) / 345	93.5 (0.49) / 669	89.1 (0.76) / 405
ACI ($\gamma=0.05$)	92.4 (1.16) / 718	92.1 (0.67) / 822	86.6 (1.14) / 405 ^{2.8}	92.5 (0.58) / 609 ^{11.8}	89.8 (0.75) / 446
Transport+ACI ($\gamma=0.02$)	92.6 (1.07) / 713	91.1 (0.80) / 787	86.5 (1.17) / 437 ^{0.6}	92.4 (0.60) / 986 ^{1.1}	90.3 (0.73) / 459
Transport+ACI ($\gamma=0.05$)	92.0 (1.19) / 681	90.9 (0.81) / 806	87.4 (1.13) / 446 ^{9.9}	91.6 (0.63) / 718 ^{10.7}	90.5 (0.72) / 489
Base-model CRPS (MWh)	96.1	133.5	70.5	94.5	79.0

The value of the method is not that it predicts curtailment better in absolute terms, but that it attaches to each prediction an interval that remains informative even as the system changes regime. This is what a decision-maker who manages a portfolio of assets actually needs.

Three threats to validity deserve emphasis. First, exchangeability is violated even within a single regime, because the daily systemic state is shared across plants and the scores arrive in correlated daily blocks; the finite-sample guarantee of the static split is therefore marginal and approximate, and this is why every coverage figure is reported with a date-clustered standard error. Second, coverage differences of one to three points between methods lie within that standard error, and we refrain from over-interpreting small gaps. Third, because a decision-maker evaluates each plant individually, conditional reliability matters: a method that is sharper on average but less reliable on the largest plants, which carry the greatest economic weight, is not necessarily preferable.

Several directions remain open. A change-point detector could trigger recalibration at the onset of a regime shift rather than adapting continuously, which would address a coverage dip observed in mid-2025 that no method anticipates. A heteroscedastic base model with a conditional dispersion $\sigma(x)$ was tested and yielded sharper intervals on average but degraded conditional coverage on the largest plants; the diagnosis is that the storage shift acts on the location rather than on the dispersion of the predictive, so modelling the variance cannot correct a shift in the mean,

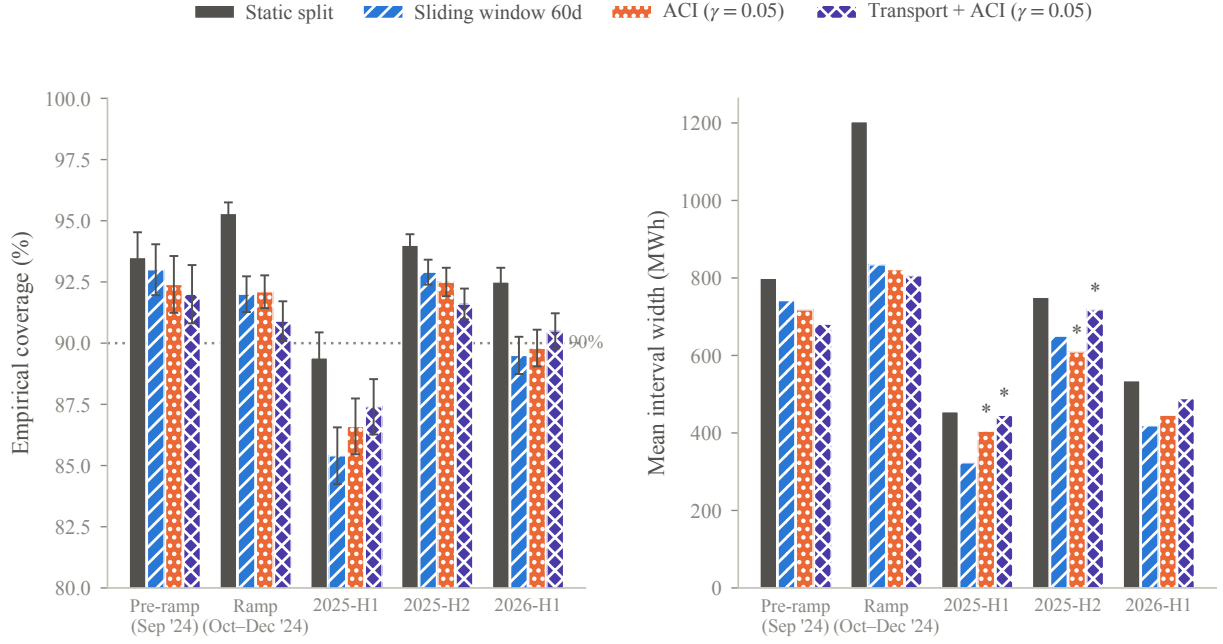


Figure 4: Coverage and sharpness by method and period. Left: empirical coverage with date-clustered standard errors against the 90% nominal level. Right: mean interval width. Asterisks mark bars whose mean width is computed over finite intervals only, because the corresponding configuration returns at least 1% of infinite intervals. The adaptive schemes reduce width during the ramp while holding coverage near nominal.

and we retain the constant-dispersion model for operational reliability. Extending the transport-based calibration to the multivariate setting, across plants that share a transmission corridor, is a natural next step.

7. Conclusion

We have presented an open, errata-corrected, plant-level dataset of renewable curtailment for Chile’s National Electricity System and a probabilistic forecasting method that pairs a two-part model with a conformal calibration layer designed to remain informative under regime change. The empirical study shows that point forecasting offers little marginal value for this problem, that a storage-driven regime shift inflates standard conformal intervals into over-covered and wider bands rather than breaking their coverage, that the textbook remedy for covariate shift fails on a controlled panel by returning uninformative intervals, and that an adaptive scheme based on a probability-integral-transform score with online recalibration and score transport restores sharpness through the transition while holding coverage near nominal. The dataset and code are publicly available, and we hope both the resource and the method are useful to operators, asset owners and researchers working on the integration of variable renewable energy.

CRedit authorship contribution statement

Pablo Reyes Cerda: Conceptualization, Data curation, Software, Validation, Investigation, Writing – original draft. **Kerven Cea Morales:** Methodology, Formal analysis, Validation, Writing – review and editing.

Declaration of competing interest

Pablo Reyes Cerda is the founder of Synapta SpA and of CurtailmentIQ, a commercial forecasting service that uses data sources related to those described in this article. Kerven Cea Morales declares no competing interest.

Declaration of generative AI in scientific writing

During the preparation of this work the authors used an AI-based assistant to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the article.

Data availability

The dataset is openly available on Zenodo under a CC BY 4.0 licence (DOI: 10.5281/zenodo.21198816, which resolves to the latest version of the deposit). The processing pipeline, the modelling code and the scripts that regenerate every table and figure in this article are available in the companion repository at <https://github.com/Pablosebastian-reyes/curtailmentiq-research>.

Appendix A. Inversion of the PIT limit

Fix x and write $p = p(x)$, $\mu = \mu(x)$. The mixture CDF is $F_x(y) = 0$ for $y < 0$, $F_x(0) = 1 - p$, and $F_x(y) = (1 - p) + p \Phi((\log y - \mu)/\sigma)$ for $y > 0$, which is continuous and strictly increasing on $(0, \infty)$ from $1 - p$ to 1. Given a level $\hat{q} \in [0, 1]$ we seek the smallest $u \geq 0$ with $F_x(u) \geq \hat{q}$. If $\hat{q} \leq 1 - p$ the atom already reaches the level, so $u = 0$. If $\hat{q} \geq 1$ no finite u attains it and $u = +\infty$. For $1 - p < \hat{q} < 1$, strict monotonicity gives a unique solution of $(1 - p) + p \Phi((\log u - \mu)/\sigma) = \hat{q}$, that is $\Phi((\log u - \mu)/\sigma) = (\hat{q} - (1 - p))/p \in (0, 1)$, hence $\log u = \mu + \sigma \Phi^{-1}((\hat{q} - (1 - p))/p)$ and

$$U(x) = \exp\left(\mu + \sigma \Phi^{-1}\left(\frac{\hat{q} - (1 - p)}{p}\right)\right),$$

which is (3). The two boundary cases are recovered as the limits of the clipped argument $(\hat{q} - (1 - p))/p \rightarrow 0^+$ and $\rightarrow 1^-$, since $\Phi^{-1}(0^+) = -\infty$ gives $U = 0$ and $\Phi^{-1}(1^-) = +\infty$ gives $U = +\infty$. $\square$

Appendix B. Coverage under score transport

Static split. If the calibration scores $s_1, \dots, s_n$ and the test score s_{n+1} are exchangeable, then with $\hat{q} = s_{(k)}$, $k = \lceil (n + 1)(1 - \alpha) \rceil$, one has $\mathbb{P}(s_{n+1} \leq \hat{q}) \geq k/(n + 1) \geq 1 - \alpha$, and since F_x is monotone, $s_{n+1} \leq \hat{q}$ if and only if $Y \leq U(x)$; hence $\mathbb{P}(Y \leq U(x)) \geq 1 - \alpha$. This is a *marginal* statement, and it is exact only under exchangeability; the serial and cross-sectional dependence of the panel make it approximate, which is why the reported standard errors are clustered by date.

Transported scores. Let P_{old} and P_{new} be the laws of the score in the old and new regimes and $T = F_{\text{new}}^{-1} \circ F_{\text{old}}$ the monotone transport map. Under an *oracle* map, $T(s) \sim P_{\text{new}}$ for $s \sim P_{\text{old}}$, so the transported calibration scores are exchangeable with the test score and the static guarantee transfers exactly. With an *estimated* map $\hat{T}$ the transported scores are only approximately distributed as P_{new} , and the coverage gap is bounded by the total-variation distance $d_{\text{TV}}(\hat{T}_{\#}P_{\text{old}}, P_{\text{new}})$ through the weighted-exchangeability framework of Barber et al. [4]. We do not rely on this bound in practice: because $\hat{T}$ is data-dependent and the gap is not directly controllable, we obtain validity operationally by feeding the transported scores into the adaptive update of Gibbs and Candès [5], whose long-run coverage $\frac{1}{T} \sum_t \text{err}_t \rightarrow \alpha$ holds for any sequence, so that the transport contributes sharpness and the online control contributes coverage.

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Declaration of interests

☐The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☒The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Pablo Sebastian Reyes Cerda reports a relationship with Synapta SpA that includes: employment and equity or stocks. Pablo Sebastian Reyes Cerda reports a relationship with CurtailmentIQ that includes: employment and equity or stocks. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.